

PrepPilot

An Offline-First, AI-Assisted Study and Examination Preparation Platform

Quick Report · Archit Jain · October 2026

1. What PrepPilot Is

PrepPilot helps a student organise an examination syllabus as Subject → Chapter → Topic → optional Sub-topic, tick off what is genuinely finished, time study sessions, revise, and see honest progress — with or without an internet connection. Timers, flashcards, notes, quizzes, reminders and the AI assistant all attach to that same syllabus, so effort, revision and progress stay connected instead of being spread over several apps. Its principle is **Tracker first, AI second**: the tracker is complete without the network, and the assistant can never change the syllabus without the student's confirmation.

Table 1.1: Project at a glance

Item	Details
What it is	Offline-first study tracker with optional AI help
Platform	Android app; the same code also runs as a browser demo (preppilottdemo.netlify.app)
Built with	React Native, Expo and TypeScript; SQLite with Drizzle ORM on the device
Backend	Supabase: PostgreSQL with Row Level Security, Auth, Storage, Edge Functions
AI	Google Gemini, called only through a server-side Edge Function
Content	22 bundled syllabus templates (JEE, NEET, CBSE, UPSC, GATE, CA, …)
Size and tests	About 36,000 lines of source; 2,283 automated tests

2. The Problem

Students track syllabi of hundreds of topics on paper, in spreadsheets or from memory, and switch between separate apps for timing, revision and AI help. Many study where connectivity is poor. Worse, the obvious way to compute progress — averaging chapter percentages — is wrong: a subject with a finished 1-topic chapter and an untouched 99-topic chapter shows as 50% done when the truth is 1%. PrepPilot computes progress topic-weighted and shows an empty chapter as — rather than 0%.

3. Key Features

- **Syllabus tracker** with progress rings and bars at every level, several courses, 22 templates and a Build-from-scratch form.
- **Timers:** stopwatch, Pomodoro (25/5/15) and custom, plus a full-screen landscape timer with six faces; every session is recorded against the syllabus.
- **Dashboard and sessions:** completion, today, total time, streak, last 7 days.
- **Revision:** flashcards (1d / 2d / 4d), quizzes of 1–15 questions at four difficulties saved to a Quiz deck, and any number of notes per topic.

- **AI assistant:** explain, summarise, quiz, answer questions about a photo, and suggest syllabus additions that the student reviews before they are added.
- **Focus Mode** (1–240 min) and student-created reminders.
- **Downloadable themes:** Classic built in; Notebook, Flight deck and Bold bento downloaded only when chosen.
- **Accounts and sync:** email, Google and guest sign-in; automatic cloud sync.

4. How It Works

The SQLite database on the phone is the source of truth: every action is saved there first, so nothing waits for the network. A sync engine then pulls changes from Supabase, merges them by fixed rules — a deletion wins, a topic's tick is resolved on its own timestamp so a rename elsewhere cannot undo it, otherwise the later edit wins — and pushes the result. It runs on sign-in, every five minutes, on leaving the app and on reconnecting. Row Level Security in PostgreSQL keeps each student's data private. The Gemini key exists only on the server, and the assistant's suggestions can only add chapters or topics, never delete or complete anything.

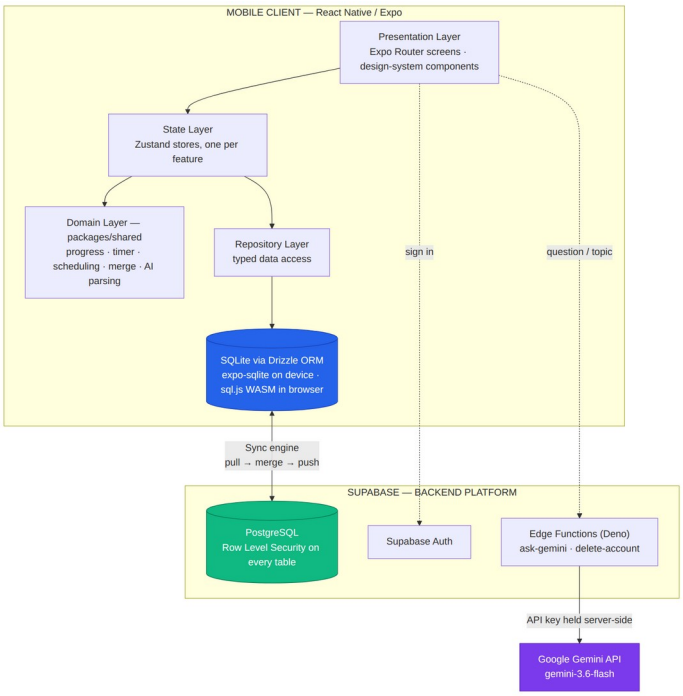


Figure 4.1: System architecture

5. Screens

Screens marked “captured 24 Sep 2026” are from the repository; those marked “web demo, 6 Oct 2026” were captured from the browser build, which uses sample data.

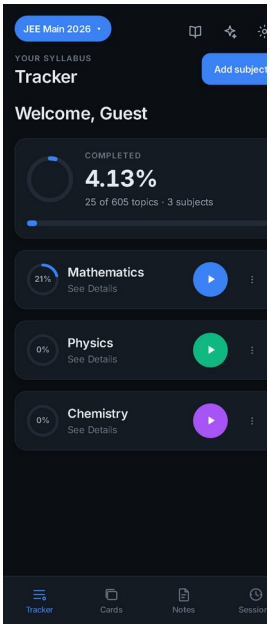


Figure 5.1: Tracker
captured 24 Sep 2026

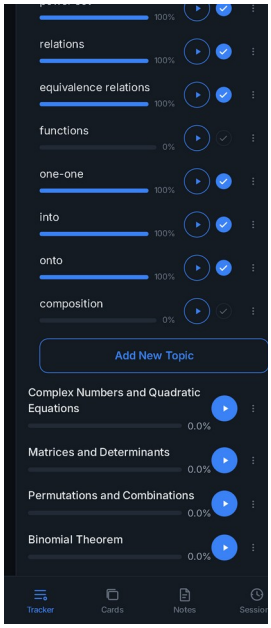


Figure 5.2: Subject expanded into chapters
captured 24 Sep 2026

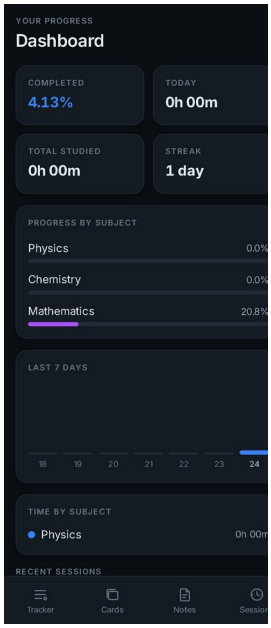


Figure 5.3: Dashboard
captured 24 Sep 2026

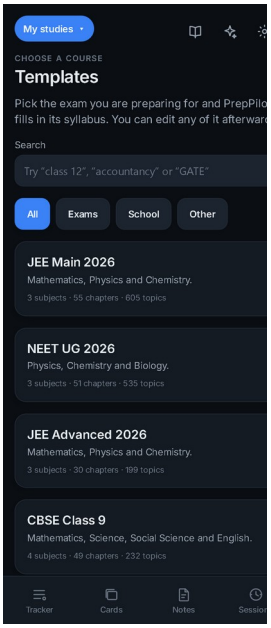


Figure 5.4: Template library
captured 24 Sep 2026

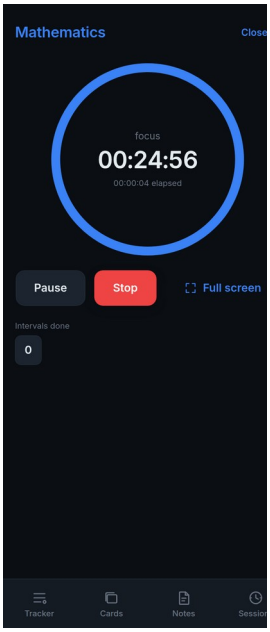


Figure 5.5: Pomodoro timer
web demo, 6 Oct 2026

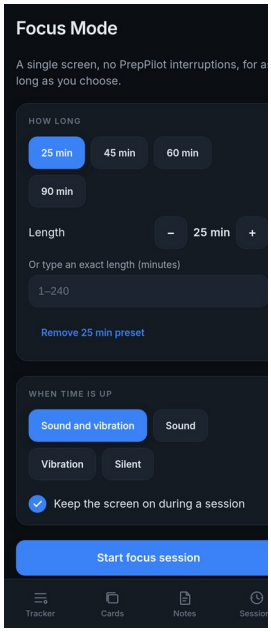


Figure 5.6: Focus Mode set-up
web demo, 6 Oct 2026

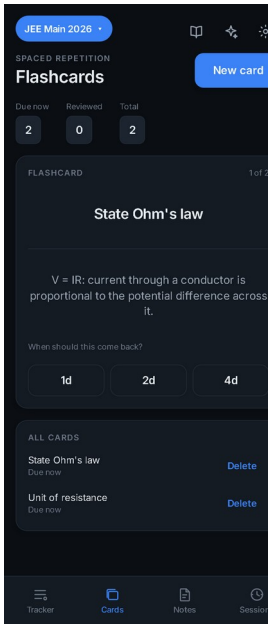


Figure 5.7: Flashcard review
captured 24 Sep 2026

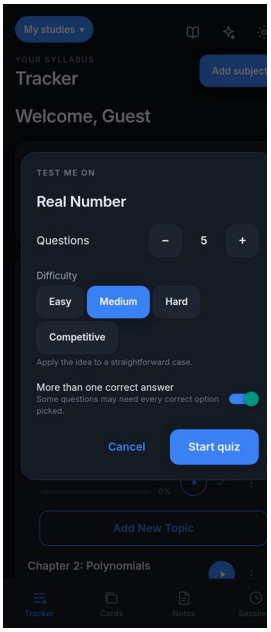


Figure 5.8: Quiz settings
web demo, 6 Oct 2026

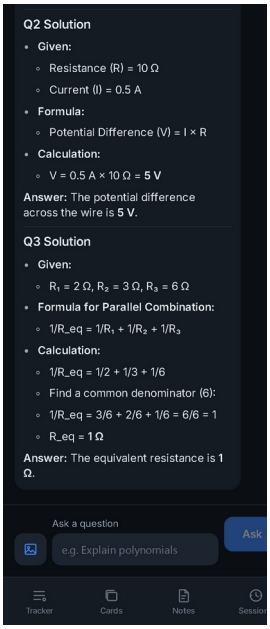


Figure 5.9: Assistant answering about a photo
captured 24 Sep 2026

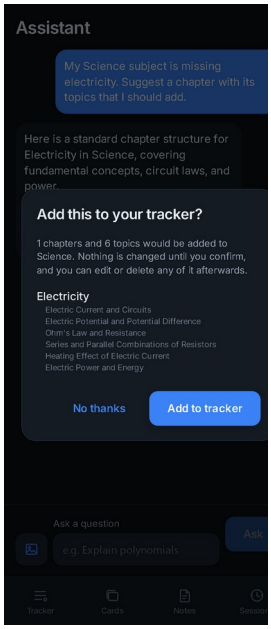


Figure 5.10: Reviewing an AI suggestion
captured 24 Sep 2026

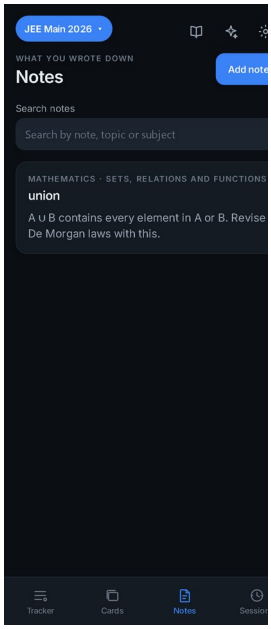


Figure 5.11: Notes
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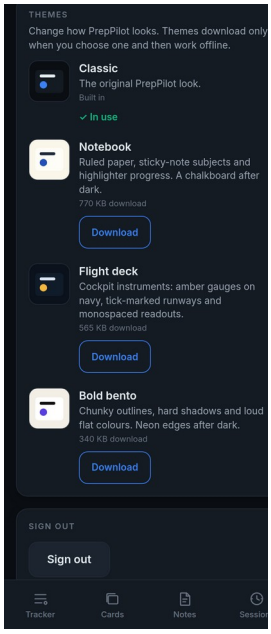


Figure 5.12: Themes in Settings
web demo, 6 Oct 2026

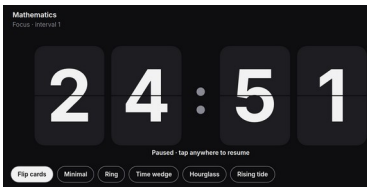


Figure 5.13: Full-screen timer: Flip cards
web demo, 6 Oct 2026

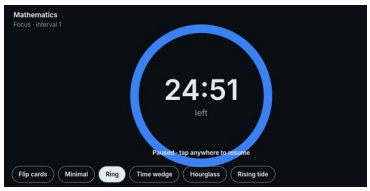


Figure 5.14: Ring face
web demo, 6 Oct 2026

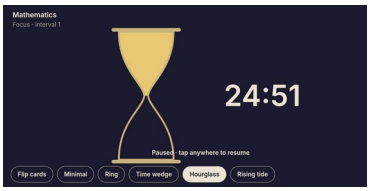


Figure 5.15: Hourglass face
web demo, 6 Oct 2026

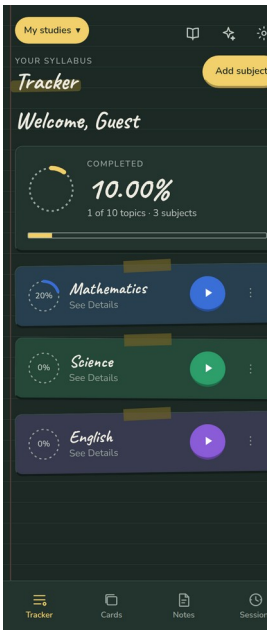


Figure 5.16: Notebook theme
web demo, 6 Oct 2026

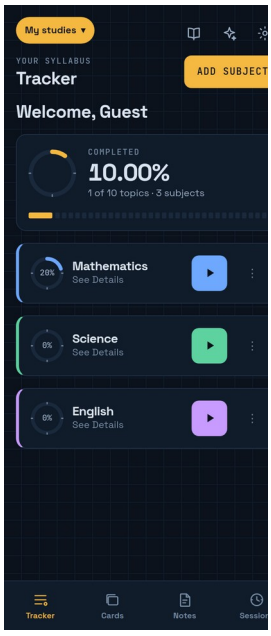


Figure 5.17: Flight deck theme
web demo, 6 Oct 2026

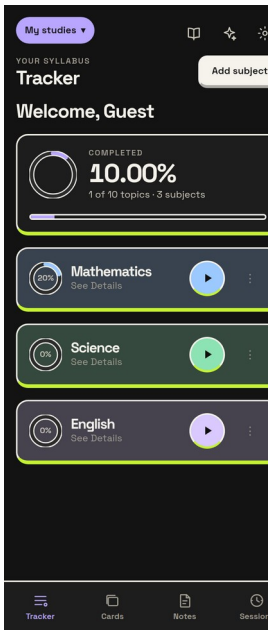


Figure 5.18: Bold bento theme
web demo, 6 Oct 2026

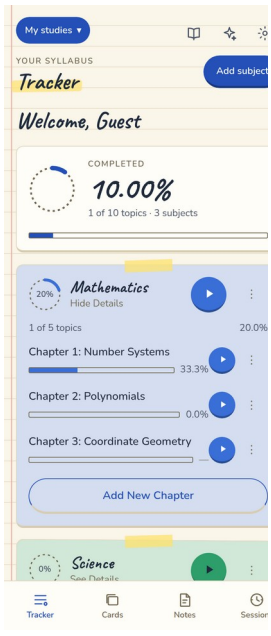


Figure 5.19: Notebook, light mode
web demo, 6 Oct 2026

6. Testing and Results

Table 6.1: Automated tests

Package	Tests	Result
App (Jest)	1,658 in 86 suites	All passed, run 6 Oct 2026
Shared logic (Vitest)	573 in 17 files	All passed, run 6 Oct 2026
Edge Functions (Deno)	52 in 2 files	Counted from source; run in CI

Continuous integration checks formatting and types, enforces coverage thresholds (95% statements and lines, 90% branches), and confirms the Android and web bundles build. Database tests run the app's real migrations against SQLite, and the sync engine is tested against an in-memory server that behaves like Supabase. According to the project documentation, the Android APK (about 41 MB) has been tested on a physical phone against the live backend, including Google sign-in, restoring data after a reinstall and photo questions. All four project objectives — accurate measurement, offline-first data, safe AI and one integrated workflow — are met.

7. Limitations and Next Steps

- Flashcards use fixed intervals; SM-2 statistics are kept so adaptive scheduling can be switched on later.
- Focus Mode cannot block other apps or turn on Do Not Disturb — Android does not allow it — and the app says so.
- The local database is not encrypted, and the AI rate limit is held in memory.
- Gemini runs on the free tier; a real launch needs the paid tier.
- Next: adaptive flashcards, local encryption, a durable rate limit, wider device testing and a Play Store release.

8. Conclusion

PrepPilot delivers a study tracker that works offline, measures progress honestly and adds AI help without giving up control of the student's work. It runs on Android and in the browser, and is backed by more than two thousand automated tests. The full Detailed Project Report covers the design, implementation and testing in depth.